

Postprandial plasma adiponectin decreases after glucose and high fat meal and is independently associated with postprandial triacylglycerols but no...



Adiponectin is discussed to regulate energy balance and insulin sensitivity. Several studies indicated an association of fasting adiponectin with parameters of the metabolic syndrome. We investigated postprandial adiponectin release and its relation to traits of the metabolic syndrome. Serum adiponectin concentration after an oral glucose tolerance test and after ingestion of a standardised mixed, fat-containing meal in 110 male non-diabetic subjects was assessed. Fasting and postprandial adiponectin and the decrease of adiponectin were correlated with anthropometric and metabolic parameters. Subjects were genotyped for adiponectin - 11 388 G/A promoter single nucleotide polymorphism. Adiponectin slightly decreased after both test meals. A significant decrease was attained 5 and 6 h after the lipid load and 2 h after the glucose load. Particularly, the mixed meal postprandial adiponectin showed stronger correlations with most traits of the metabolic syndrome than fasting adiponectin: postprandial adiponectin with HDL (r 0.30) v. fasting adiponectin with HDL (r 0.23); with postprandial insulin (area under the curve): r - 0.20 v. r - 0.16; with fasting insulin: r 0.10 v. r 0.14; with BMI: r - 0.23 v. r - 0.20; with waist: r - 0.18 v. - 0.16; with systolic blood pressure: r - 0.14 v. r - 0.12; with diastolic blood pressure: r - 0.18 v. r - 0.15. In multivariate analysis, postprandial TAG were the only independent predictor of adiponectin. There was no significant association of adiponectin, NEFA and TAG with - 11 388 G/A adiponectin promoter polymorphism. Our findings favour the interpretation that postprandial adiponectin has the strongest and independent associations to postprandial TAG metabolism.